

Ralph David Fernando

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Education & Honors

Indiana University

Bachelor of Arts: Mathematics & Economics

Hutton Honors Scholar (Maintain 3.7 GPA)

Dean's List (Earned at least 3.7 GPA in College of Arts & Sciences)

Robert A Buccieri Jr Scholarship (Presented to an Economics major who has demonstrated academic excellence throughout their studies)

Bloomington, IN

Expected Graduation: **May 2025**

August 2021 - Present

(Jan 2021, May 2022, Jan 2022)

April 2023

Coding Languages

- Python, R, Stata, Java, C++, Basic
- GitHub Repository: <https://github.com/ralphx34/Ralph-F.-Repo.git>

Skills

- Excel, LaTeX, Nvivo, Qualtrics

Research Experience

Economics Department, Indiana University - Research Assistant

(Summer 24) Bloomington, IN

- Analyzed experimental economics research papers relevant to framing in prisoner's dilemma and public goods games
- Created bibliography using aforementioned papers for use in writing a literature review
- Devised tables and documents using Overleaf for use in experimental economics presentations
- Formulated large dataset of economics questions designed to study ChatGPT's responses

Political Science Department, Indiana University - Research Assistant

(Summer 24) Bloomington, IN

- Built datasets on small businesses that qualified for Covid-19 Assistance Programs in Excel
- Organized USDA county-level SNAP data using Python
- Collected data on House candidates from 2018-2022 general elections
- Using 2022 Cooperative Election Study data, constructed tables for analyzing perceptions on economic well-being

Mathematics Department, Indiana University, Lab of Geometry - Student Participant

(Fall 23) Bloomington, IN

- Introduced to finite difference schemes and optimization in numerical analysis
- Analyzed machine learning algorithms and physics-informed neural networks
- Used the Pytorch package in Python, coded machine learning algorithms to solve various physics informed PDE's
- Applied acquired knowledge to code a machine learning algorithm for the Black-Scholes equation

IU Undergraduate Advanced Summer Research Project - Student Researcher

(Summer 23) Bloomington, IN

- Acquired understanding of 'Text as Data' methodology
- Became familiar with quantitative data practices in Political Science and Sociology
- Created a best practices guideline on different 'Text as Data' research processes
- Used Nvivo to code interviews done by the research team on the hardships of Russian emigrants at the beginning of the conflict with Ukraine

Arts & Science Undergraduate Research Experience - Student Participant (Fall 21-Spring 22) Bloomington, IN

- Learned how to think critically and logically in a project-based learning environment
- Began and developed a unique research project that was presented in a professional manner
- Gained familiarity with data collection techniques and the academic research process

Work Experience

Everything Policy, Content Research & Analysis Intern (Sept 24 - Pres)

- Conduct independent research on numerous political topics such as healthcare costs and border policing
- Use mathematical and statistical models to analyze patterns
- Work and communicate with a supervisor to present data and information
- Prepare summaries, abstracts, and visual representations of data that provide insights

IU Economics Department, Lead Tutor (Aug 23 - Pres)

- Act as a liaison between professors and department tutors
- Relay administrative information to fellow peer tutors
- Guide other tutors on potential conflicts or issues that arise
- Assist new tutors with any potential difficulties

IU Economics Undergraduate Tutor (Aug 22 - Pres)

- Manage tutoring sessions held by the department of Economics
- Serve as additional academic support to attending students
- Work with students in a 1-on-1 setting
- Build up my own ability to illustrate concepts in a clear and concise manner

IU Economics Undergraduate Intern (UGI) (Jan 22 - Pres)

- Provide support to the instructor during lecture and assist with grading
- Hold office hours and provide additional assistance to 200+ students
- Conduct collaborative learning sessions to further students' understanding of class material
- Proctor exams including all midterms and finals
- Explain various information to students, boosting proficiency

Classes Assisted:

1. Introductory Microeconomics (1 semester)
2. Introductory Macroeconomics (3 semesters)
3. Intermediate Microeconomics (3 semesters)
4. Honors Intermediate Microeconomics (1 semester)